



SecuredXWave

CYBER SAFETY SIMPLIFIED

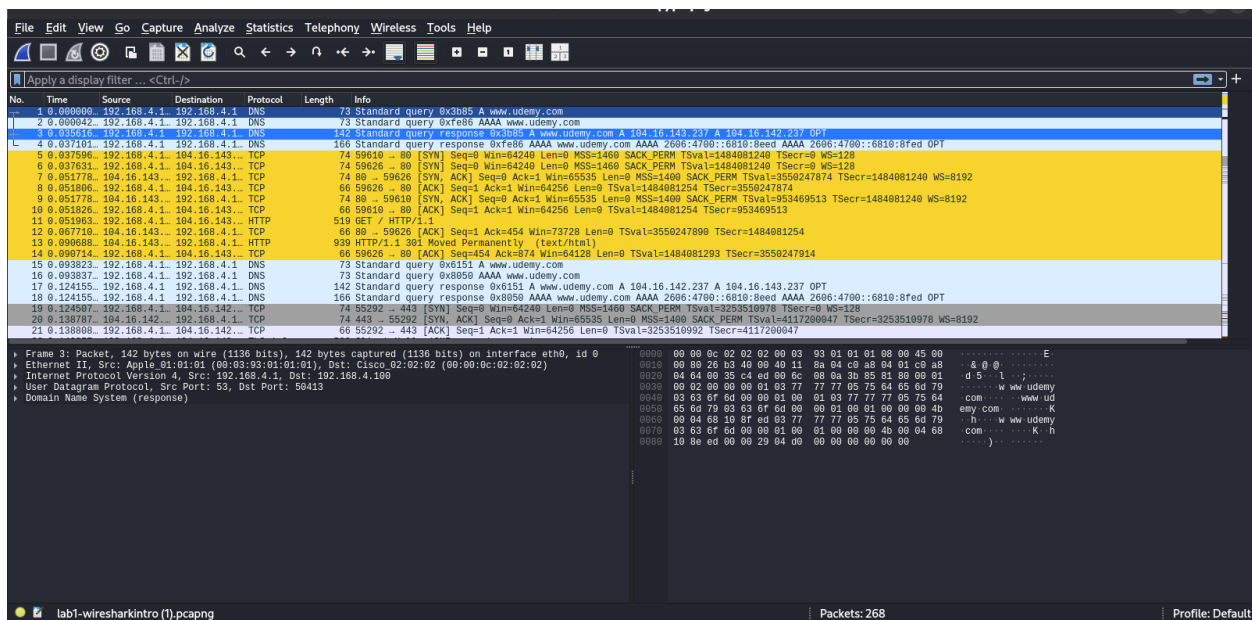
Day 5: Packet analysis theory · Wireshark filters · How traffic is captured

Section A — Getting Oriented

Question No 1 :

Open the file and identify the three main panes of the Wireshark window. What does each one show?

I opened the pcap file and saw three panes of the wireshark window in my kali linux.



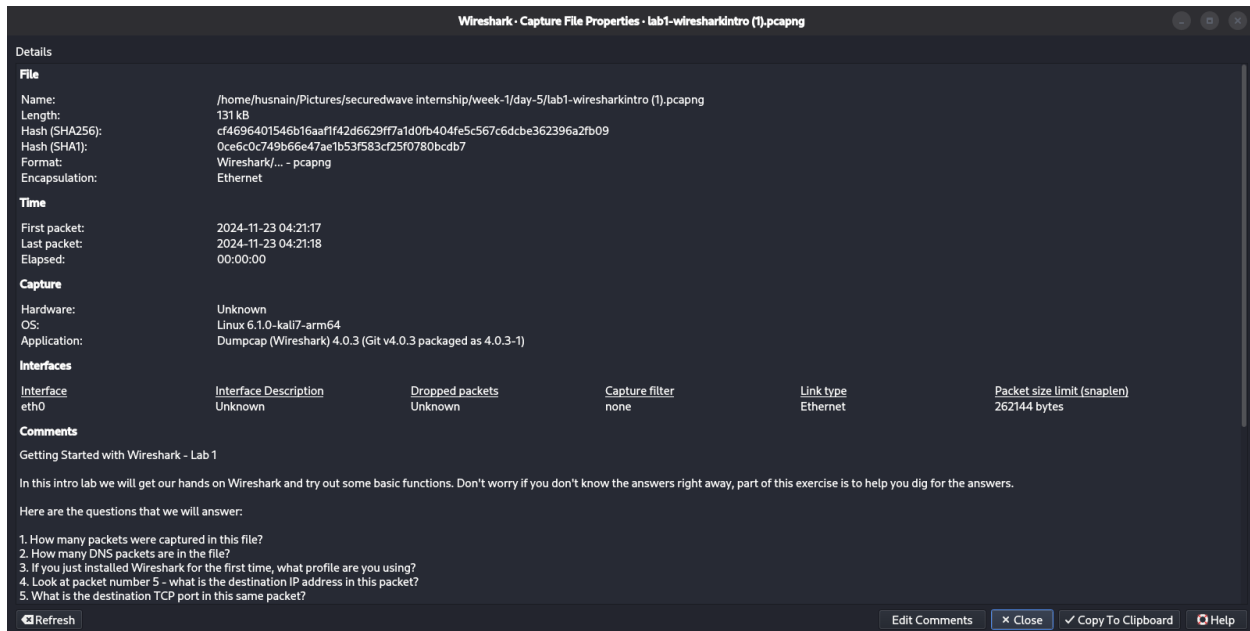
- There are 3 panes in the wireshark window.
- First, top pane shows the Time, Source IP address, Destination IP address, Protocols, Length of packets and Info about the packets
- Second, bottom left pane shows the details of selected packet and expand each protocol layer to inspect the packet.

- Third, bottom right pane shows the actual data of the packet in the hexadecimal and ASCII

Question No 2:

How many packets are in this capture, and roughly how long did the capture last?

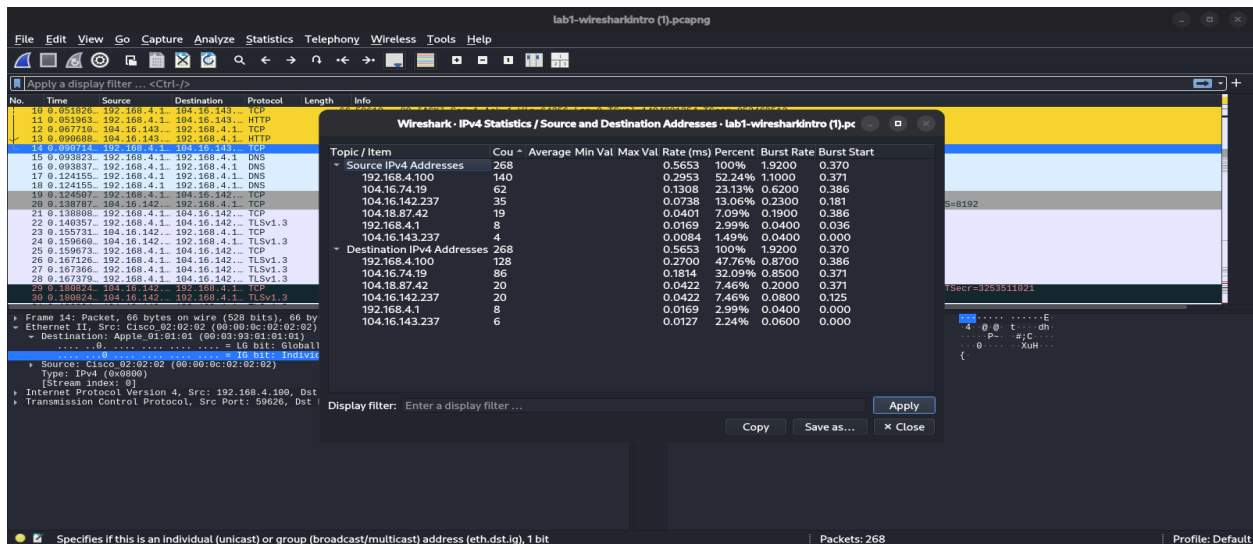
- I go to the Statistics section on the top bar and select the Capture File Properties.
- There was mentioned that there is **268 packets** in this capture file. The capture lasted **approximately half a second** and I verify it using the timestamps:
First packet: 2024-11-23 04:21:17
Last packet: 2024-11-23 04:21:18
- Since those timestamps only show whole seconds, they look like a 1-second difference. However, Wireshark stores timestamps with fractions of a second, and the Time span (0.474 s) is the precise capture duration.



Question No 3:

What IP address belongs to "our" machine (the one doing the browsing)?

- Opened the statistics section from top bar, selected ipv4 statistics -> source and destination ipv4 statistics.
- So there is an IP **192.168.4.100** that must belong to "our" machine because it initiates DNS queries and TCP connections and appears as both the source and destination in most conversations.

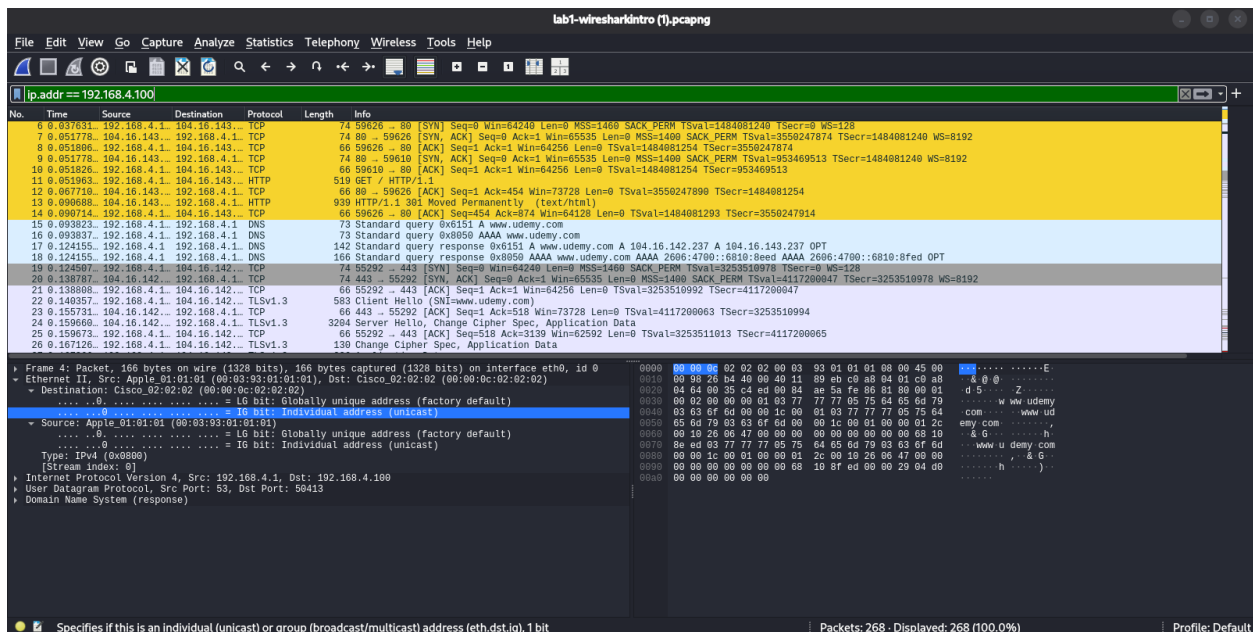


Section B — Filtering Basics

Question No 4:

Type `ip.addr == 192.168.4.100` in the display filter bar. What happens, and why is this useful?

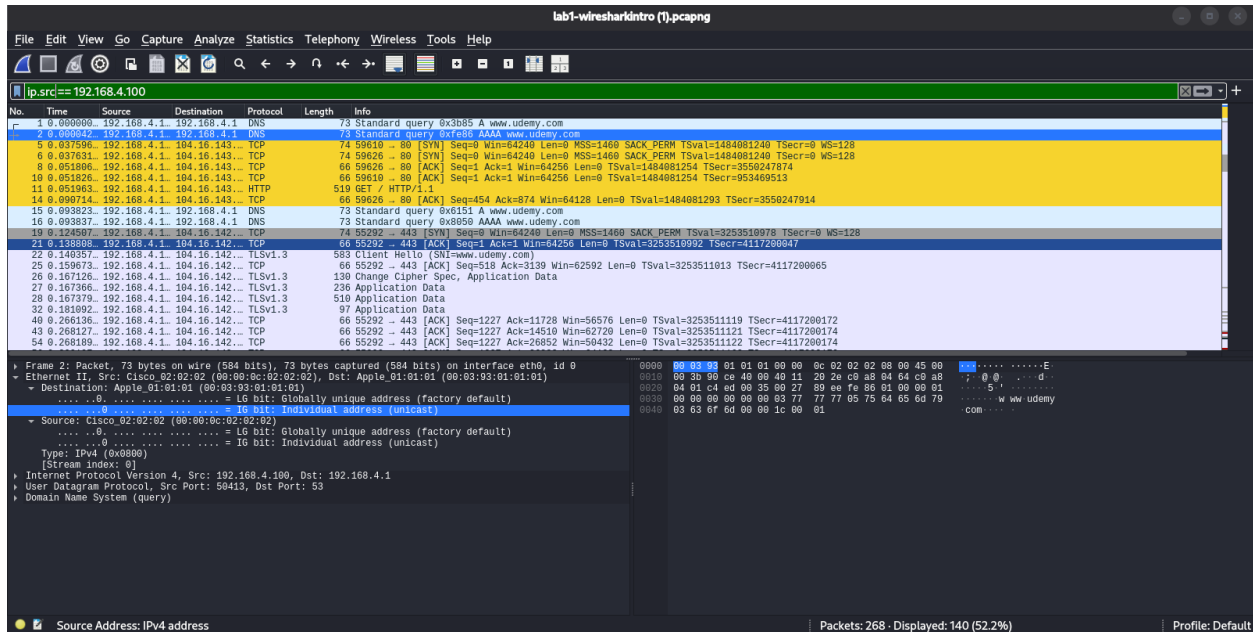
- When i type `ip.addr == 192.168.4.100` in the filter section then it **shows only that specific IP traffics**.
- It shows the both source and destination packets of this IP.
- It is useful for filtering and specific IP packets.
- 268 packets was displayed while filtering.



Question No 5:

What's the difference between `ip.addr == 192.168.4.100` and `ip.src == 192.168.4.100`?

- The difference between `ip.addr == 192.168.4.100` and `ip.src == 192.168.4.100` is that **ip.addr** matches every packet where 192.168.4.100 either it is Source IP or Destination IP appears while **ip.src** only matches packets where 192.168.4.100 is the sender (source).
- When i apply `ip,.src` filter then it displayed 1440 packets that's mean filter work correctly and shows only that packets where 192.168.4.100 is sender.



Question No 6:

Filter for `tcp.port == 443`. What do you see, and what does port 443 tell you about the traffic?

- After applying the filter `tcp.port == 443`, I observed TCP and TLSv1.3 packets, including the Client Hello, Server Hello, Change Cipher Spec, and encrypted Application Data packets.
- Port 443 indicates that the traffic is HTTPS, meaning the communication between the client and server is encrypted using TLS.

- File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.4.1	192.168.4.1	DNS	73	Standard query 0x3b85 A www.udemy.com
2	0.000042	192.168.4.1	192.168.4.1	DNS	142	Standard query 0xfef8 AAAA www.udemy.com
3	0.000516	192.168.4.1	192.168.4.1	DNS	142	Standard query response 0x3b85 A www.udemy.com A 194.16.143.237 A 194.16.142.237 OPT
4	0.037101	192.168.4.1	192.168.4.1	DNS	166	Standard query response 0xfef8 AAAA www.udemy.com AAAA 2606:4700:6810:beed AAAA 2606:4700:6810:8fed OPT
5	0.038303	192.168.4.1	192.168.4.1	DNS	73	Standard query 0x6151 A www.udemy.com
6	0.093837	192.168.4.1	192.168.4.1	DNS	73	Standard query 0x8059 AAAA www.udemy.com
7	0.124155	192.168.4.1	192.168.4.1	DNS	142	Standard query response 0x6151 A www.udemy.com A 194.16.142.237 A 194.16.143.237 OPT
8	0.124155	192.168.4.1	192.168.4.1	DNS	166	Standard query response 0x8059 AAAA www.udemy.com AAAA 2606:4700:6810:beed AAAA 2606:4700:6810:8fed OPT
9	0.338225	192.168.4.1	192.168.4.1	DNS	82	Standard query 0xe676 A frontends.udemycdn.com
10	0.338241	192.168.4.1	192.168.4.1	DNS	82	Standard query 0x1075 AAAA frontends.udemycdn.com
11	0.339804	192.168.4.1	192.168.4.1	DNS	77	Standard query 0x1fbf A cdn.cookieiaw.org
12	0.339100	192.168.4.1	192.168.4.1	DNS	77	Standard query 0xfefb AAAA cdn.cookieiaw.org
13	0.360285	192.168.4.1	192.168.4.1	DNS	169	Standard query response 0xe676 A frontends.udemycdn.com A 194.16.74.19 A 194.16.75.19 OPT
14	0.370255	192.168.4.1	192.168.4.1	DNS	170	Standard query response 0xfefb AAAA cdn.cookieiaw.org AAAA 2606:4700:6812:562a AAAA 2606:4700:6812:572a OPT
15	0.370255	192.168.4.1	192.168.4.1	DNS	154	Standard query response 0x1fbf A cdn.cookieiaw.org A 194.18.87.42 A 194.18.86.42 OPT
16	0.370255	192.168.4.1	192.168.4.1	DNS	193	Standard query response 0x1075 AAAA frontends.udemycdn.com AAAA 2606:4700:6810:4b13 AAAA 2606:4700:6810:4b13 OPT

```

Frame 10: Packet, 166 bytes on wire (1328 bits), 166 bytes captured (1328 bits) on interface eth0, id 0
  Ethernet II, Src: Apple_01:01:01 (00:03:93:01:01:01), Dst: Cisco_02:02:02 (00:00:0c:02:02:02)
    Destination: Cisco_02:02:02 (00:00:0c:02:02:02)
      ... = 10 bit: Globally unique address (factory default)
      ... = 10 bit: Individual address (unicast)
    Source: Apple_01:01:01 (00:03:93:01:01:01)
      ... = 10 bit: Globally unique address (factory default)
      ... = 10 bit: Individual address (unicast)
    Type: IPv4 (0x0806)
    Stream index: 91
  Internet Protocol Version 4, Src: 192.168.4.1, Dst: 192.168.4.100
  User Datagram Protocol, Src Port: 53, Dst Port: 54417
  Domain Name System (response)
  
```

☒ Specifies if this is an individual (unicast) or group (broadcast/multicast) address (eth.dst.ig), 1 bit

Packets: 268 - Displayed: 16 (6.0%)

Profile: Default

Section C — DNS

Question No 8:

With the dns filter applied, list every hostname this machine looked up.

- There are 3 hostnames when applying dns filter,
 - www.udemy.com
 - frontends.udemycdn.com
 - cdn.cookieclaw.org

No.	Time	Source	Destination	Protocol	Length	Info
1	0.009090	192.168.4.1	192.168.4.1	DNS	73	Standard query 0x3b85 A www.udemy.com
2	0.009042	192.168.4.1	192.168.4.1	DNS	73	Standard query 0xfe86 AAAA www.udemy.com
3	0.03616	192.168.4.1	192.168.4.1	DNS	142	Standard query response 0x3b85 A www.udemy.com A 104.16.143.237 A 104.16.142.237 OPT
4	0.037101	192.168.4.1	192.168.4.1	DNS	166	Standard query response 0xfe86 AAAA www.udemy.com AAAA 2606:4700::6818:8eed AAAA 2606:4700::6818:8fed OPT
15	0.093823	192.168.4.1	192.168.4.1	DNS	73	Standard query 0x8151 A www.udemy.com
16	0.093837	192.168.4.1	192.168.4.1	DNS	73	Standard query 0x8050 AAAA www.udemy.com
17	0.124155	192.168.4.1	192.168.4.1	DNS	142	Standard query response 0x8151 A www.udemy.com A 104.16.142.237 A 104.16.143.237 OPT
18	0.242150	192.168.4.1	192.168.4.1	DNS	166	Standard query response 0x8050 AAAA www.udemy.com AAAA 2606:4700::6818:8eed AAAA 2606:4700::6818:8fed OPT
58	0.338225	192.168.4.1	192.168.4.1	DNS	82	Standard query 0xe876 A frontends.udemycdn.com
59	0.338241	192.168.4.1	192.168.4.1	DNS	82	Standard query 0x1075 AAAA frontends.udemycdn.com
60	0.338994	192.168.4.1	192.168.4.1	DNS	77	Standard query 0x17bf A cdn.cookieclaw.org
61	0.339100	192.168.4.1	192.168.4.1	DNS	77	Standard query 0x7bf7 AAAA cdn.cookieclaw.org
67	0.369285	192.168.4.1	192.168.4.1	DNS	169	Standard query response 0xe876 A frontends.udemycdn.com A 104.16.74.19 A 104.16.75.19 OPT
68	0.370255	192.168.4.1	192.168.4.1	DNS	178	Standard query response 0x7bf7 AAAA cdn.cookieclaw.org AAAA 2606:4700::6812:562a AAAA 2606:4700::6812:572a OPT
69	0.370255	192.168.4.1	192.168.4.1	DNS	154	Standard query response 0x17bf A cdn.cookieclaw.org A 104.18.87.42 A 104.18.86.42 OPT
70	0.370255	192.168.4.1	192.168.4.1	DNS	193	Standard query response 0x1075 AAAA frontends.udemycdn.com AAAA 2606:4700::6818:4a13 AAAA 2606:4700::6818:4b13 OPT

Question No 9:

Pick one DNS response packet. What IP address did the server return for www.udemy.com?

- I choosed www.udemy.com and The DNS server returned the IPv4 addresses 104.16.143.237 and 104.16.142.237 for www.udemy.com.

```
... ..0. .... = LG bit: Globally unique address (factory default)
... ..0. .... = IG bit: Individual address (unicast)
Source: Apple_01:01:01 (00:03:93:01:01:01)
... ..0. .... = LG bit: Globally unique address (factory default)
... ..0. .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 192.168.4.1, Dst: 192.168.4.100
User Datagram Protocol, Src Port: 53, Dst Port: 50413
Domain Name System (response)
Transaction ID: 0x3b85
Flags: 0x8180 Standard query response, No error
Questions: 1
Answer RRs: 2
Authority RRs: 0
Additional RRs: 1
Queries
Answers
www.udemy.com: type A, class IN, addr 104.16.143.237
www.udemy.com: type A, class IN, addr 104.16.142.237
Additional records
[Request In: 1]
[Time: 35.616333 milliseconds]
```

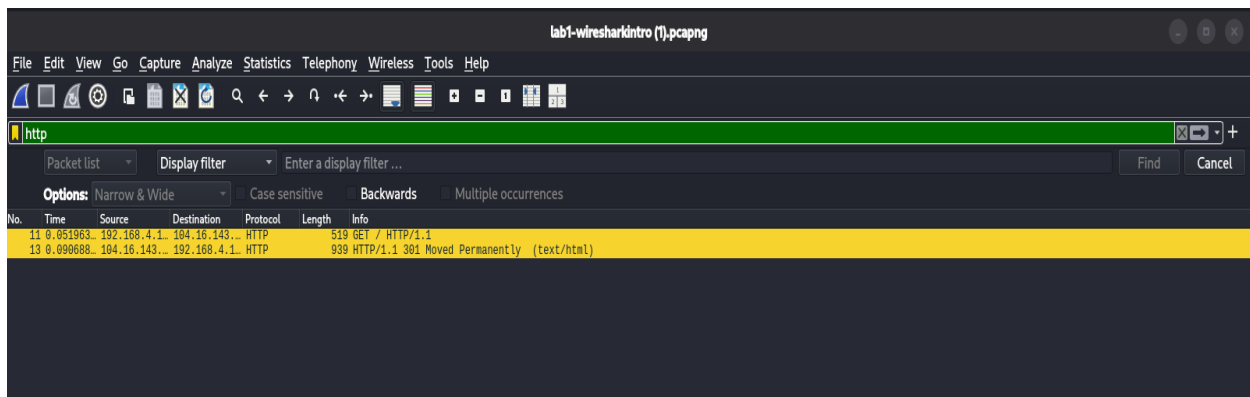
Domain Name System: Protocol

Section D — HTTP

Question No 10:

Filter for http. What do you find?

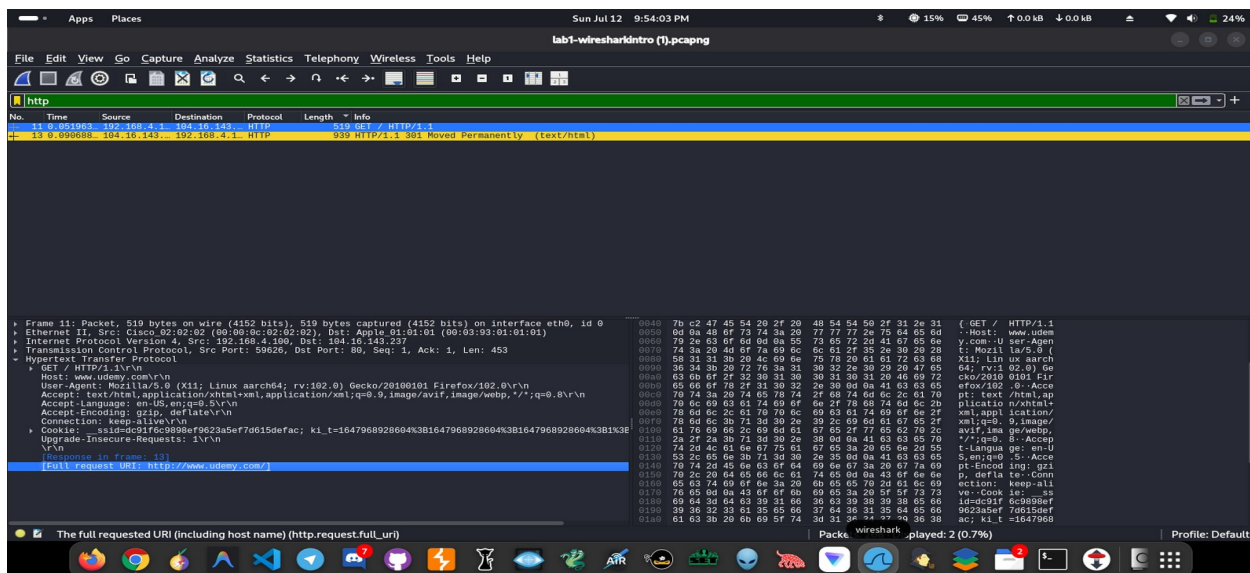
- When i apply http filter then it shows an HTTP GET request from the client and an HTTP/1.1 301 Moved Permanently response from the server.



Question No 11:

Select the GET request packet and expand it fully. What can you read in plain text that you could NOT read if this were HTTPS?

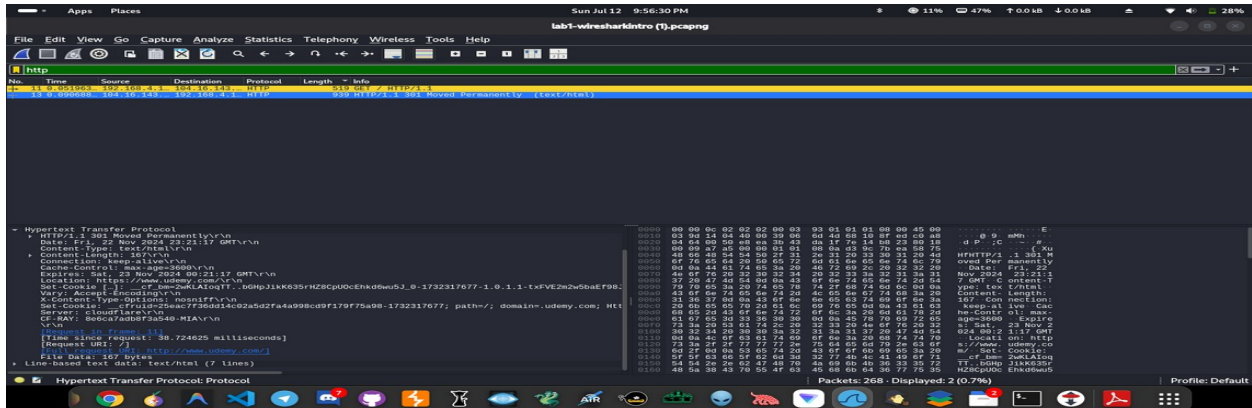
- The HTTP GET request is sent in plain text. I can read the requested URL, host name (www.udemy.com), browser information (User-Agent), and HTTP headers. If this were HTTPS, this information would be encrypted and not directly readable.



Question No 12:

Look at the server's response to that GET request. What status code does it return, and what does that mean?

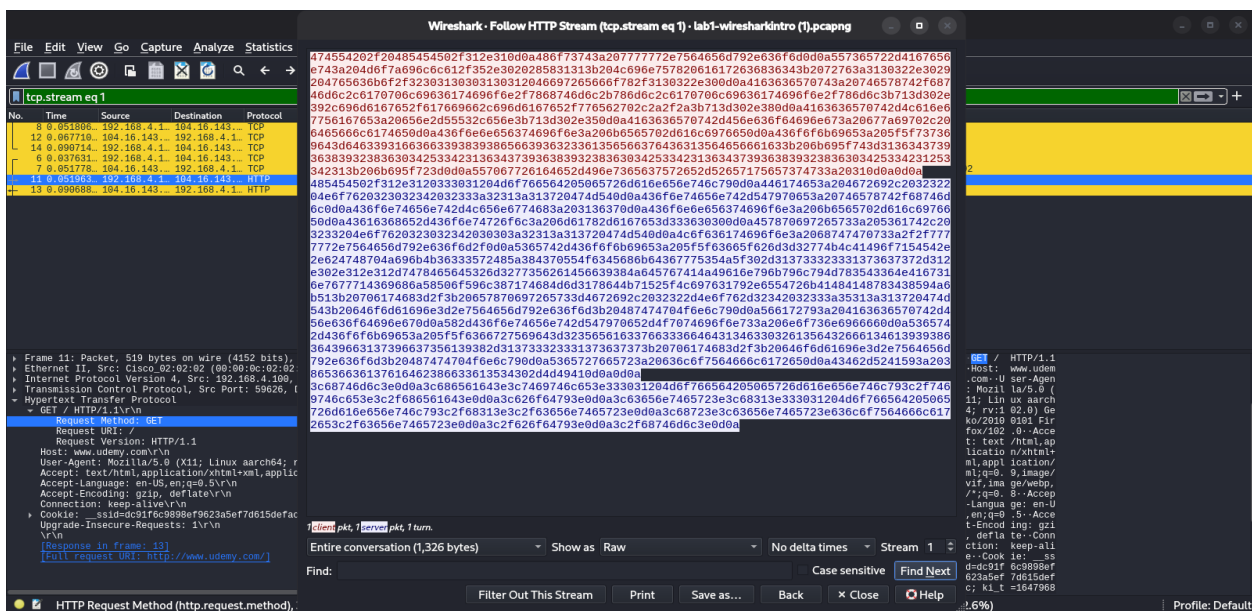
- The server returns HTTP Status Code 301 (Moved Permanently). It redirects the browser from HTTP to the secure HTTPS version of the website.



Question No 13:

Right-click the HTTP GET packet → Follow → HTTP Stream. What's different about this view compared to the packet list?

- The HTTP Stream view combines all HTTP packets into one complete conversation between the client and the server. It is easier to read than viewing individual packets because the entire request and response are shown together.

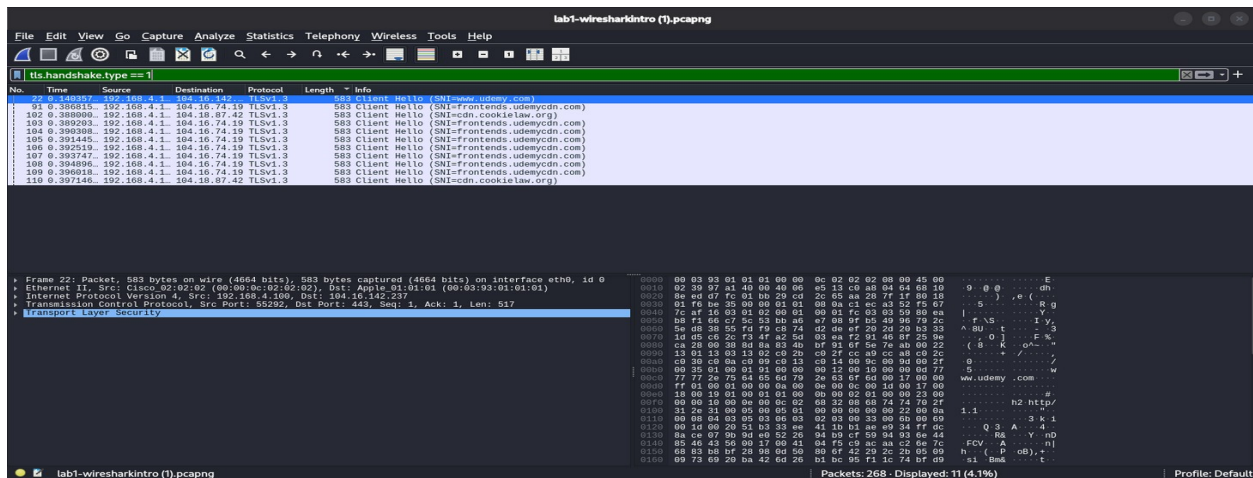


Section E — TLS/HTTPS

Question No 14:

Filter for `tls.handshake.type == 1` (Client Hello). Open one and find the Server Name Indication (SNI). What does it tell you, given the packet is otherwise encrypted?

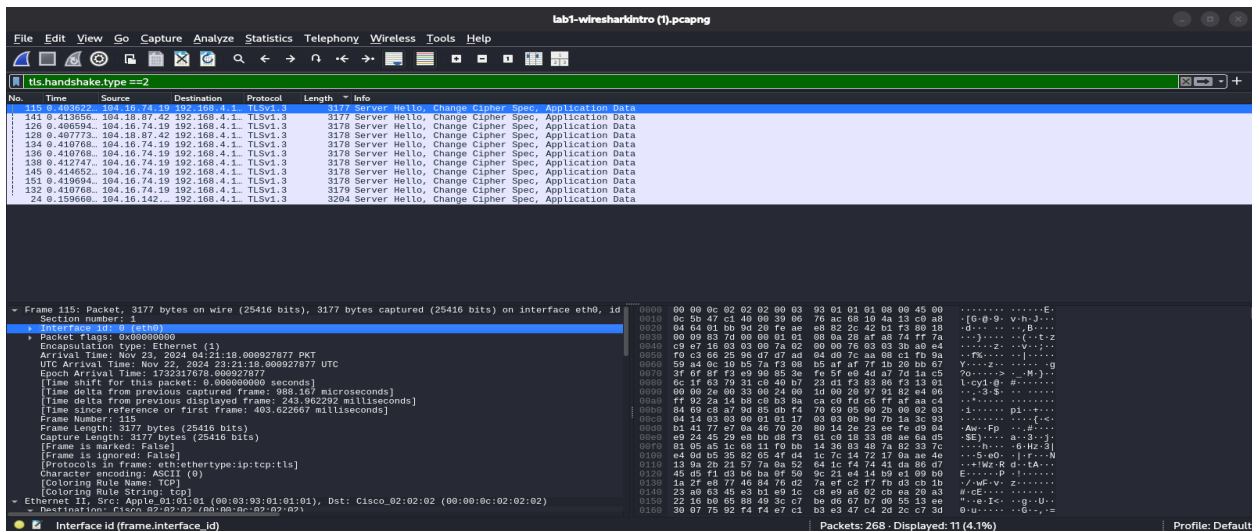
- The **Server Name Indication** (SNI) is `www.udemy.com`. Although the TLS session encrypts the communication, the SNI remains visible during the Client Hello, allowing the server to know which website the client wants to connect to before encryption is established.



Question No 15:

Filter for `tls.handshake.type == 2` (Server Hello). What is the server telling the client in this packet?

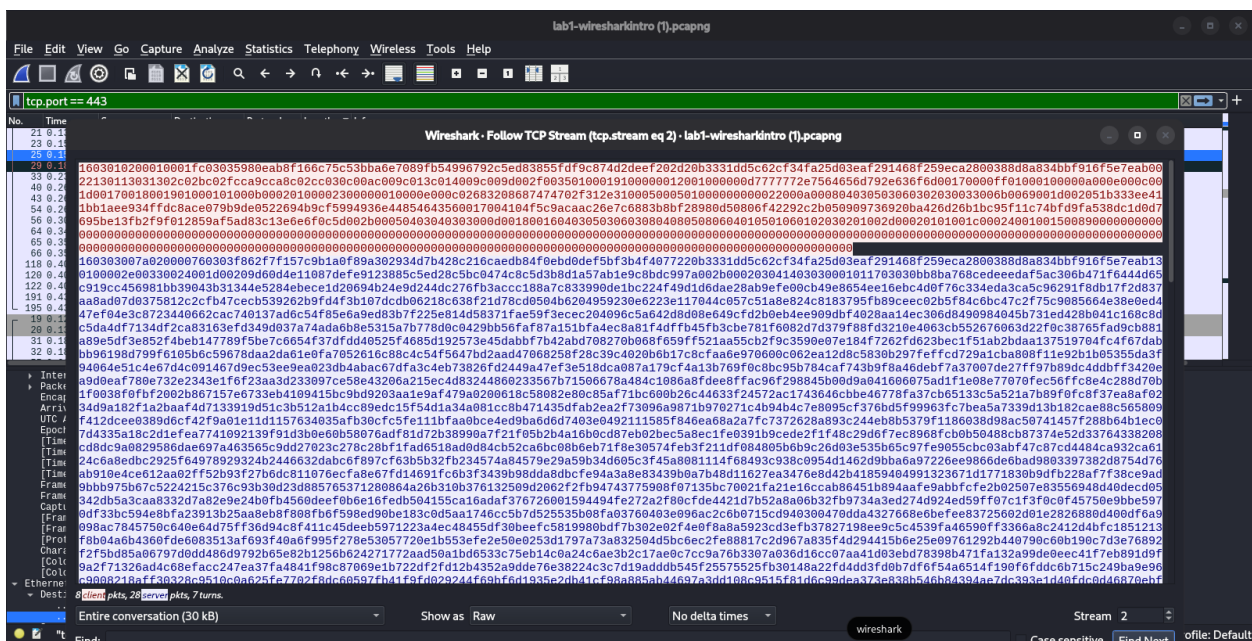
- The Server Hello tells the client which TLS version, cipher suite, and security parameters have been selected for the secure connection. This confirms how the encrypted communication will be established.



Question No 16:

Follow → TCP Stream on one of the port-443 conversations. What do you see, and why?

- When following the TCP stream on port 443, the data appears as unreadable or encrypted binary data. This is because HTTPS uses TLS encryption, which protects the contents of the communication from being read by anyone intercepting the traffic.

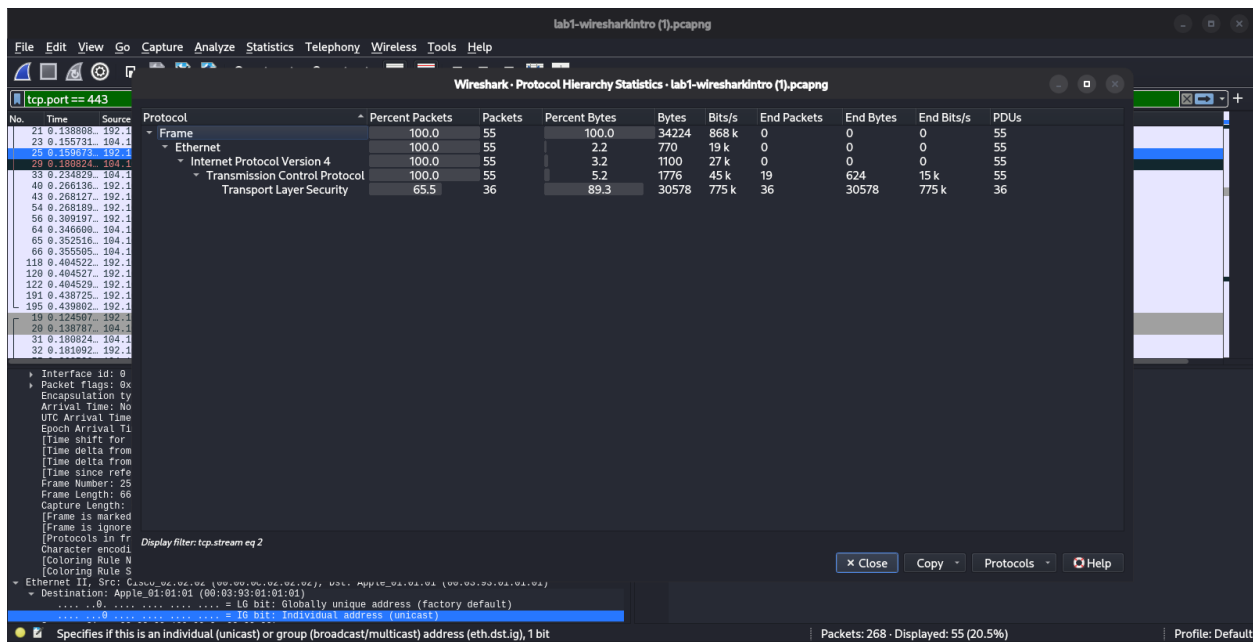


Section F — Statistics Menu

Question No 17:

Open Statistics → Protocol Hierarchy. What does this tell you about the overall makeup of the capture?

- The Protocol Hierarchy shows that the capture mainly consists of Ethernet, IPv4, TCP, and TLS traffic. Most of the communication is TLS-encrypted, with 36 out of 55 packets (65.5%) using TLS and 89.3% of the transmitted bytes being encrypted, indicating that the connection is primarily secure HTTPS traffic.



Question No 18:

Open Statistics → Conversations (IPv4 tab). Which remote host had the most packets exchanged with 192.168.4.100, and what does that suggest?

- The remote host with the most packets exchanged with 192.168.4.100 is 104.16.74.19 (148 packets). This suggests that it was the primary server the client communicated with during the capture, indicating most of the web traffic and data transfer occurred with this host.

Wireshark · Conversations · lab1-wiresharkintro (1).pcapng

Conversation Settings		IPv4 · 1	TCP · 13	UDP · 4					
☐ Name resolution		Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A
☐ Absolute start time		192.168.4.100	104.16.74.19	148	60 kB	4	86	14	0
☐ Display raw data		192.168.4.100	104.16.142.237	55	34 kB	2	20	3	0
☐ Limit to display filter		192.168.4.100	104.16.143.237	10	2 kB	1	6	865 bytes	0
		192.168.4.100	104.18.87.42	39	20 kB	3	20	3	0
		192.168.4.100	192.168.4.1	16	2 kB	0	8	610 bytes	0

Question No 19:

Open Statistics → Endpoints. How many unique IP addresses does 192.168.4.100 talk to in this whole capture?

- 192.168.4.100 communicates with 5 unique IP addresses during the capture: one local router/DNS server (192.168.4.1) and four remote internet hosts (104.16.74.19, 104.16.142.237, 104.16.143.237, and 104.18.87.42).

Wireshark · Endpoints · lab1-wiresharkintro (1).pcapng

Endpoint Settings		Ethernet · 2	IPv4 · 6	IPv6	TCP · 17	UDP · 5					
☐ Name resolution		Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Count		
☐ Display raw data		104.16.74.19	0	0 bytes	0	0 bytes	0	0 bytes	0		
☐ Hide aggregated		104.16.142.237	55	34 kB	35	31 kB	20	3 kB	0		
☐ Limit to display filter		104.16.143.237	0	0 bytes	0	0 bytes	0	0 bytes	0		
		104.18.87.42	0	0 bytes	0	0 bytes	0	0 bytes	0		
		192.168.4.1	0	0 bytes	0	0 bytes	0	0 bytes	0		
		192.168.4.100	55	34 kB	20	3 kB	35	31 kB	0		

Copy Map

Protocol

- ☒ Ethernet
- ☐ FC
- ☐ FDDI
- ☐ IEEE 802.11
- ☐ IEEE 802.15.4
- ☐ ILNP
- ☒ IPv4
- ☒ IPv6
- ☐ IPX
- ☐ JXTA

Filter list for specific type

Close Help

Wrap-up Discussion

Question No 20:

Walk through the full sequence of events in order: DNS lookup → TCP handshake → HTTP request → redirect → TLS handshake → encrypted data. Can you reconstruct this timeline using the packet numbers alone?

- Yes. The packet sequence shows the complete communication process:
- DNS Lookup – The client queries the DNS server to resolve `www.udemy.com` and receives its IP address.
- TCP Three-Way Handshake – The client and server establish a TCP connection using `SYN → SYN-ACK → ACK`.
- HTTP Request – The client sends an HTTP GET request on port 80.
- HTTP Redirect – The server replies with 301 Moved Permanently, redirecting the client to HTTPS.
- TLS Handshake – The client and server exchange Client Hello and Server Hello messages to establish a secure session.
- Encrypted Data – After the handshake, all application data is encrypted and exchanged over port 443.

Question No 21:

Why might a network admin still tell a user visited "udemy.com" even though the connection is HTTPS?

- Because the domain name is still visible through DNS queries and the TLS Server Name Indication (SNI) in the Client Hello packet. Although the website content is encrypted, the destination domain (`www.udemy.com`) can still be identified.

Question No 22:

How would you actually decrypt the TLS traffic in Wireshark if you needed to?

- TLS traffic can be decrypted by obtaining the TLS session keys (SSL Key Log File) from the client browser or application and loading the key log file into Wireshark (Edit → Preferences → Protocols → TLS → (Pre)-Master-Secret log filename). Once the correct session keys are provided, Wireshark can decrypt and display the HTTPS traffic.